



Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE)

(An Autonomous Institute)

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TY B.Tech

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Semster:5

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Assignment 3

Title: Design a Dynamic Web Application like Fashion Store / E-Commerce Product using HTML, CSS & JavaScript (Bootstrap CDN if required)

Objectives:

- To understand the development of a dynamic web application using HTML, CSS, and JavaScript.
- To design an e-commerce style interface for displaying fashion products.
- To implement interactivity using JavaScript (product display, buttons, dynamic actions).
- To apply Bootstrap components for responsive and modern UI design.
- To learn client-side scripting for improving user experience.
- To simulate basic e-commerce functionalities such as product listing, add-to-cart interaction, and responsive layouts.

Outcomes:

- Students will be able to create a dynamic web interface using JavaScript.
- Students will understand how front-end technologies work together in web application development.
- Students will gain experience designing responsive e-commerce layouts.
- Students will learn DOM manipulation and event handling using JavaScript.
- Students will be able to design user-friendly product pages similar to real online shopping platforms.

Theory Concept:

A dynamic web application is a website that updates content automatically based on user interaction without reloading the entire page. In an e-commerce fashion store, dynamic behavior improves usability by allowing users to browse products, view details, and interact with items smoothly.

1. HTML (Structure Layer)

HTML provides the basic structure of the e-commerce website. It defines elements such as product cards, navigation bar, product sections, images, and buttons.

2. CSS (Presentation Layer)

CSS is used to style the application by controlling layout, colors, fonts, spacing, and visual appearance. CSS improves readability and enhances the shopping experience.

3. JavaScript (Behavior Layer)

JavaScript makes the website dynamic by adding interactivity such as:

- Displaying products dynamically
- Add-to-cart functionality
- Button click events
- Updating cart count
- DOM manipulation without page reload
-

4. Bootstrap Framework

Bootstrap is a responsive front-end framework used through a CDN link. It provides:

- Grid system for layout alignment
- Responsive navigation bar
- Cards for product display
- Buttons and forms
- Mobile-friendly responsiveness

Using Bootstrap reduces development time and ensures consistent UI design.

5. Dynamic Web Applications in E-Commerce

Modern fashion e-commerce platforms rely heavily on dynamic interfaces to improve customer engagement. Features like product filtering, quick view, and cart updates enhance user interaction and simulate real online shopping environments.

Dataset Used:

No external dataset was used. The application uses manually created sample product data. The data is stored in JavaScript arrays/objects and dynamically rendered on the webpage.

Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>MyMart | Fashion Store</title>

  <!-- Bootstrap CDN -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css"
>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- NAVBAR -->
  <nav class="navbar bg-white shadow-sm px-4">
    <span class="navbar-brand fw-bold">MyMart</span>

    <div class="d-flex align-items-center">
      <input class="form-control me-3 search-box" placeholder="Search for
clothes">
      🛒 Cart <span id="cart-count" class="badge bg-dark ms-1">0</span>
    </div>
  </nav>

  <!-- OFFER BANNER -->
  <div class="container mt-3">
    <div class="offer-banner text-center">
      🔥 GET UP TO 50% OFF ON WINTER COLLECTION 🔥
    </div>
  </div>

  <!-- CATEGORY ICONS -->
  <div class="container mt-4">
    <div class="row g-3 text-center">

      <div class="col">
```

```

        <div class="category-card active" onclick="filterCategory('all',
this)">
            <i class="bi bi-grid-fill"></i>
            <span>All</span>
        </div>
    </div>

    <div class="col">
        <div class="category-card" onclick="filterCategory('tshirt', this)">
            <i class="bi bi-person-fill"></i>
            <span>T-Shirts</span>
        </div>
    </div>

    <div class="col">
        <div class="category-card" onclick="filterCategory('shirt', this)">
            <i class="bi bi-bag-fill"></i>
            <span>Shirts</span>
        </div>
    </div>

    <div class="col">
        <div class="category-card" onclick="filterCategory('pants', this)">
            <i class="bi bi-scissors"></i>
            <span>Pants</span>
        </div>
    </div>

    <<div class="col">
        <div class="category-card" onclick="filterCategory('shoes', this)">
            <i class="bi bi-box-seam"></i>
            <span>Shoes</span>
        </div>
    </div>

    <div class="col">
        <div class="category-card" onclick="filterCategory('watch', this)">
            <i class="bi bi-watch"></i>
            <span>Watches</span>
        </div>
    </div>

</div>
</div>

```

```
<!-- PRODUCTS -->
<div class="container mt-4">
  <div class="row g-4" id="product-list">

    <!-- T-SHIRT -->
    <div class="col-md-3 product tshirt">
      <div class="product-card">
        
        <h6>T-Shirt</h6>
        <p>₹999</p>
        <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('T-Shirt') ">
          + Add to Cart
        </button>
      </div>
    </div>

    <!-- SHIRT -->
    <div class="col-md-3 product shirt">
      <div class="product-card">
        
        <h6>Casual Shirt</h6>
        <p>₹1499</p>
        <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Shirt') ">
          + Add to Cart
        </button>
      </div>
    </div>

    <!-- PANTS -->
    <div class="col-md-3 product pants">
      <div class="product-card">
        
        <h6>Denim Pants</h6>
        <p>₹1799</p>
        <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Pants') ">
          + Add to Cart
        </button>
      </div>
    </div>

  </div>
</div>
```

```

<!-- SHOES -->
<div class="col-md-3 product shoes">
  <div class="product-card">
    
    <h6>Sports Shoes</h6>
    <p>₹2999</p>
    <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Shoes') ">
      + Add to Cart
    </button>
  </div>
</div>

<!-- WATCH -->
<div class="col-md-3 product watch">
  <div class="product-card">
    
    <h6>Wrist Watch</h6>
    <p>₹3499</p>
    <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Watch') ">
      + Add to Cart
    </button>
  </div>
</div>

<!-- MEN JACKET -->
<div class="col-md-3 product jacket">
  <div class="product-card">
    
    <h6>Men Jacket</h6>
    <p>₹1999</p>
    <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Men Jacket') ">
      + Add to Cart
    </button>
  </div>
</div>

<!-- WOMEN DRESS -->
<div class="col-md-3 product dress">
  <div class="product-card">
    
    <h6>Women Dress</h6>

```

```

                <p>₹2499</p>
                <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Women Dress')">
                    + Add to Cart
                </button>
            </div>
        </div>

        <!-- HOODIE -->
        <div class="col-md-3 product hoodie">
            <div class="product-card">
                
                <h6>Hoodie</h6>
                <p>₹1499</p>
                <button class="btn btn-outline-dark w-100"
onclick="handleAddToCart('Hoodie')">
                    + Add to Cart
                </button>
            </div>
        </div>

    </div>
</div>

<!-- LOGIN MODAL -->
<div class="modal fade" id="loginModal">
    <div class="modal-dialog">
        <div class="modal-content">
            <div class="modal-header">
                <h5>Login Required</h5>
                <button class="btn-close" data-bs-dismiss="modal"></button>
            </div>
            <div class="modal-body">
                <input id="username" class="form-control mb-2"
placeholder="Username">
                <input id="password" type="password" class="form-control mb-2"
placeholder="Password">
                <button class="btn btn-dark w-100"
onclick="login()">Login</button>
            </div>
        </div>
    </div>
</div>
</div>

```

```
    <!-- Bootstrap JS -->
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>

    <script src="script.js"></script>
</body>

</html>
```

Style.css

```
body {
    background: #f6f7fb;
}

.search-box {
    width: 250px;
}

.offer-banner {
    background: linear-gradient(to right, #ff5f6d, #ffc371);
    color: white;
    padding: 15px;
    border-radius: 12px;
    font-weight: bold;
    -webkit-border-radius: 12px;
    -moz-border-radius: 12px;
    -ms-border-radius: 12px;
    -o-border-radius: 12px;
}

.product-card {
    background: white;
    border-radius: 14px;
    padding: 15px;
    text-align: center;
    box-shadow: 0 10px 25px rgba(0,0,0,0.08);
    transition: transform 0.3s;
}

.product-card:hover {
    transform: translateY(-5px);
}
```



```
.product-card img {
  height: 200px;
  object-fit: cover;
  border-radius: 10px;
  width: 100%;
}

.row > .col button {
  border-radius: 10px;
  font-weight: 500;
}

.category-card {
  background: white;
  border-radius: 14px;
  padding: 15px 10px;
  cursor: pointer;
  box-shadow: 0 8px 20px rgba(0,0,0,0.08);
  transition: all 0.3s ease;
}

.category-card i {
  font-size: 24px;
  display: block;
  margin-bottom: 6px;
}

.category-card span {
  font-size: 14px;
  font-weight: 500;
}

.category-card:hover {
  transform: translateY(-4px);
}

.category-card.active {
  background: #212529;
  color: white;
}
```

[script.js](#)

```
let isLoggedIn = false;
let cartCount = 0;
let selectedProduct = "";

function handleAddToCart(product) {
  if (!isLoggedIn) {
    selectedProduct = product;
    let modal = new bootstrap.Modal(document.getElementById("loginModal"));
    modal.show();
  } else {
    addToCart(product);
  }
}

function login() {
  let user = document.getElementById("username").value;
  let pass = document.getElementById("password").value;

  if (user === "user" && pass === "1234") {
    isLoggedIn = true;
    addToCart(selectedProduct);
    bootstrap.Modal.getInstance(document.getElementById("loginModal")).hide();
    alert("Login Successful!");
  } else {
    alert("Invalid credentials");
  }
}

function addToCart(product) {
  cartCount++;
  document.getElementById("cart-count").innerText = cartCount;
}

function filterCategory(category, element) {
  let products = document.querySelectorAll(".product");
  let categories = document.querySelectorAll(".category-card");

  categories.forEach(cat => cat.classList.remove("active"));
  element.classList.add("active");

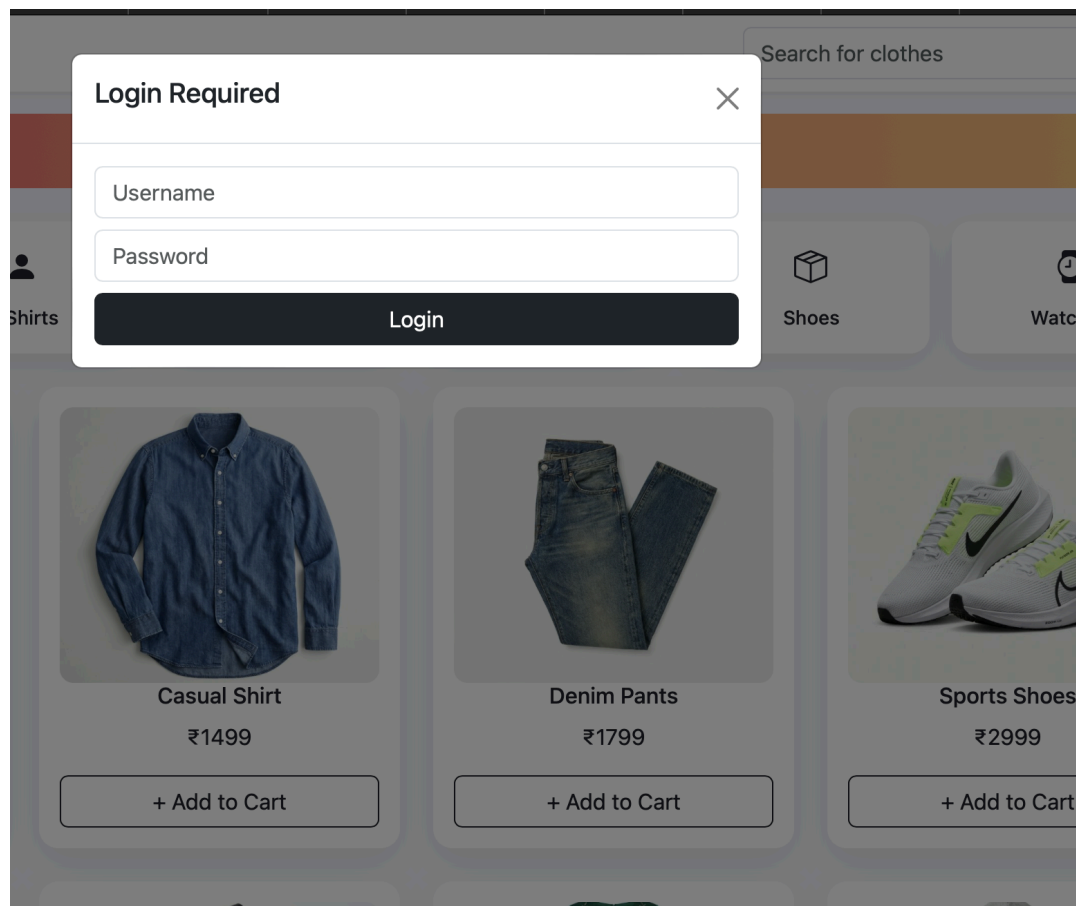
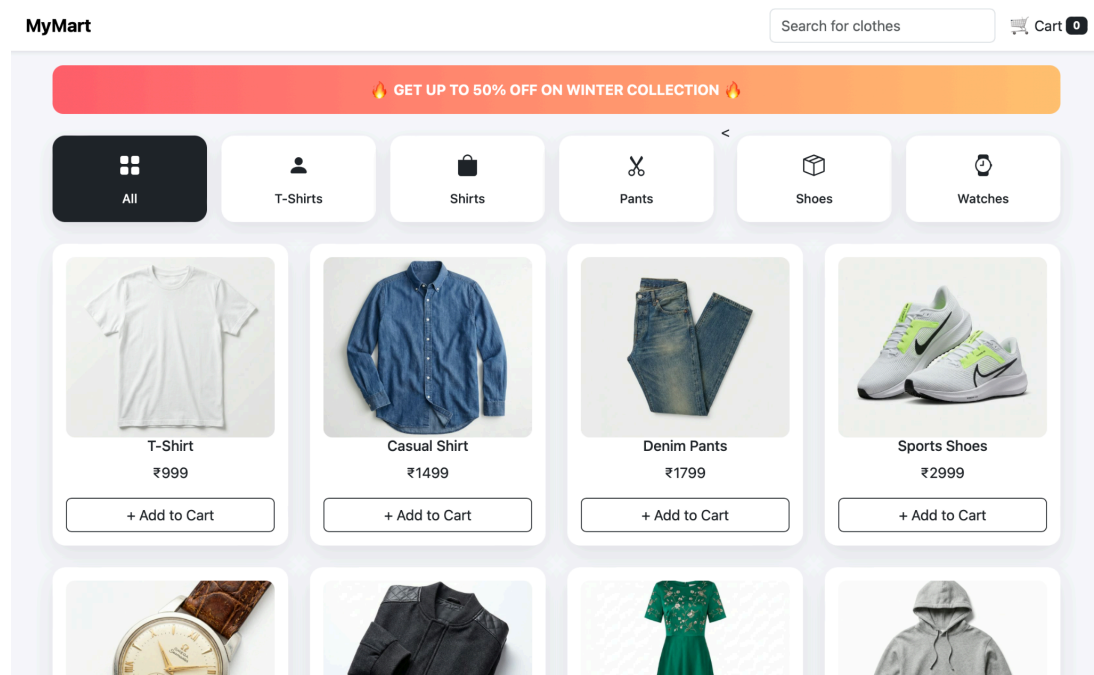
  products.forEach(product => {
    if (category === "all" || product.classList.contains(category)) {
      product.style.display = "block";
    }
  });
}
```

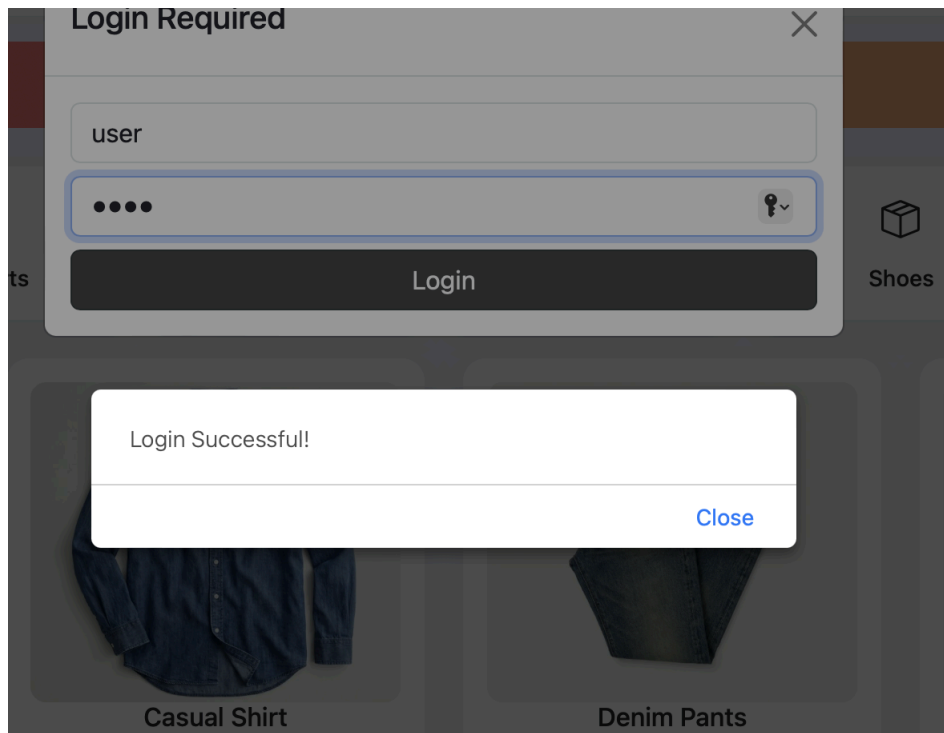
```

    } else {
        product.style.display = "none";
    }
});
}

```

Output:





Conclusion

The assignment successfully demonstrated the creation of a dynamic fashion store web application using HTML, CSS, JavaScript, and Bootstrap. The project helped in understanding client-side scripting, responsive web design, and dynamic content rendering. JavaScript enhanced user interaction by enabling real-time updates and interactive product features. This experiment strengthened practical front-end development skills essential for building modern e-commerce applications in full stack development.